

Section A

Answer **all** the questions in this section.

- 1 Enzyme **X** digests a sugar. Fig. 1.1 shows the rate of reaction of enzyme **X** and a mixture of enzyme **X** and substance **Y** on the sugar. Both reactions were carried out in a conical flask.

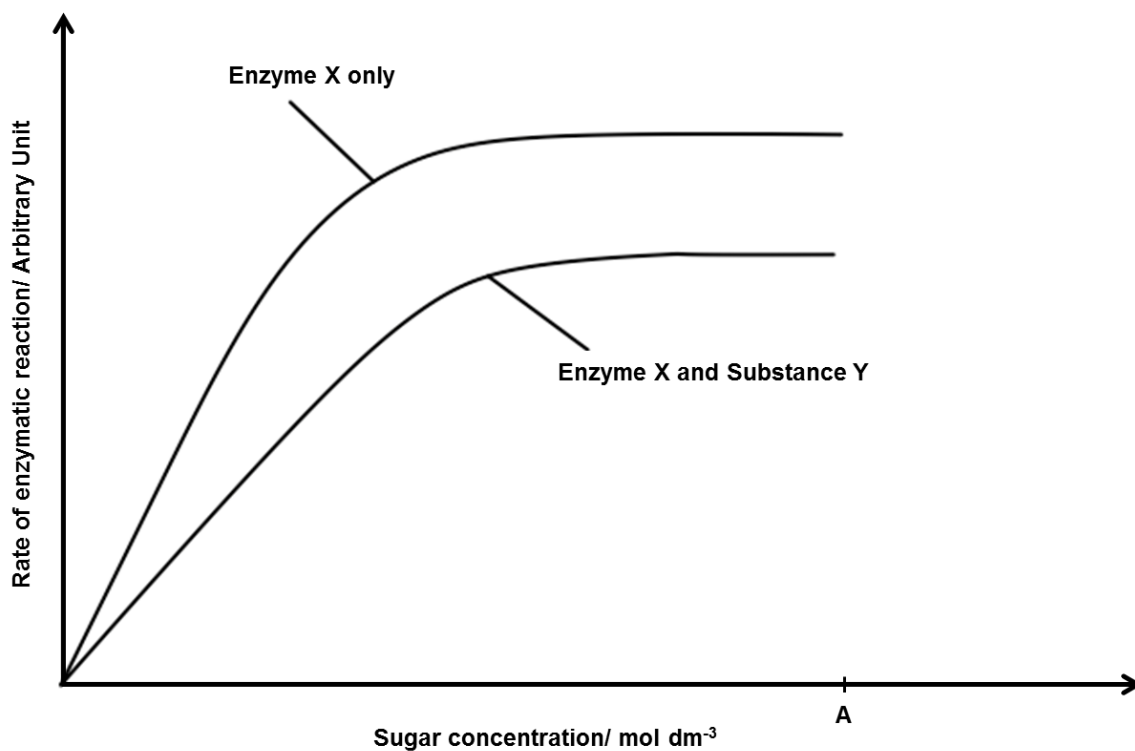


Fig. 1.1

- (a) With reference to Fig. 1.1, describe and explain the effect of substance **Y** on the enzymatic reaction. [3]

- (b) Substance **Y** was found in one of the products of the reaction catalysed by enzyme **X** after the substrate concentration indicated by **A**, in Fig.1.1.

Predict how the trend of the graph for enzyme X only would continue after sugar concentrations greater than **A**. Sketch the part of the graph in Fig. 1.1. [1]

Enzyme **X** may be immobilised in alginate beads. Fig. 1.2 shows a comparison between the activity of free enzyme **X** and immobilised enzyme **X** over a range of temperatures. Equal concentrations of free enzyme **X** and immobilised enzyme **X** were used.

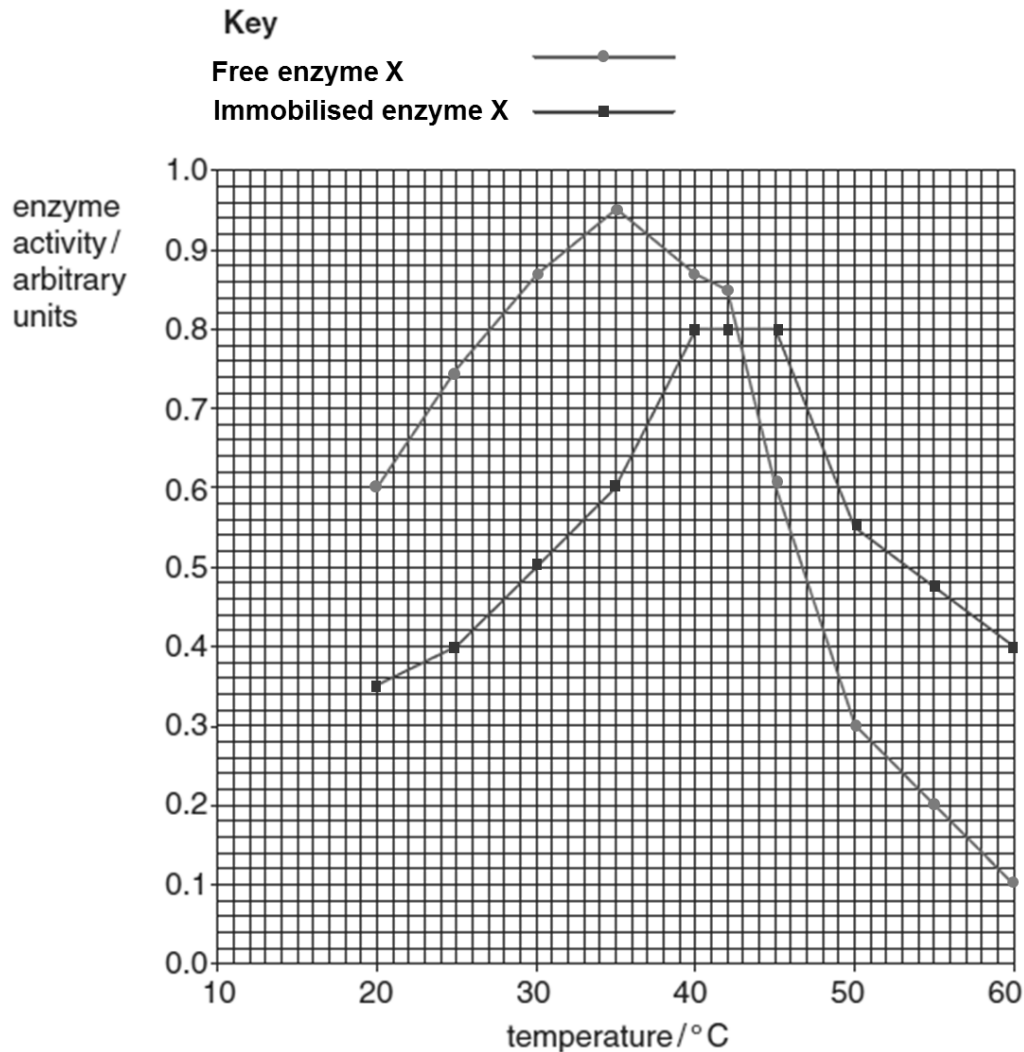


Fig. 1.2

(c) With reference to Fig. 1.2,

- (i)** describe and explain the effect of temperature on the activity of free enzyme **X**. [3]

- (ii)** describe and explain the differences between the activity of free enzyme **X** and immobilised enzyme **X** up to 40 °C. [2]

- (iii)** suggest an advantage of using immobilised enzymes instead of free enzymes. [1]

[Total: 10]

- 2 (a) Explain what is meant by homologous pairs of chromosomes. [2]

- (b) A mutant allele for eye colour was inherited in a recessive manner along with the X chromosome in the common fruit fly *Drosophila melanogaster*.

A red-eyed female fly was crossed with a white-eyed male fly and all the F_1 generation flies had red eyes. When a red-eyed female from the F_1 generation is crossed with a red-eyed male from the F_1 generation, all female flies had red eyes in the F_2 generation. Half of the male flies in the F_2 generation had white eyes and the rest of the males had red eyes.

- (i) Using suitable symbols, draw a genetic diagram to explain the results of this cross. [4]

- (ii) Explain how it can be deduced from the results of the cross that the mode of inheritance for eye colour is X-linked recessive. [2]

- (c) Observed results of the above genetic cross differ from the expected results.

Suggest **two** reasons why such a discrepancy occurs, referring only to events that occur after meiosis. [2]

[Total: 10]

- 3 An experiment was conducted to investigate the relationship between oxygen consumption and carbohydrate catabolism. A suspension of isolated mitochondria was prepared from rat liver and an intermediate of the Krebs cycle, succinic acid, was added. The reaction mixture was immediately placed in a reaction chamber of an electronic device which measured the changes in oxygen concentration in the reaction mixture. Readings were taken at 2-minute intervals for 12 minutes.

A chemical **X** was added to the reaction mixture 6 minutes after the start of the measurement. The experiment was repeated with another similarly prepared suspension of liver mitochondria but with no other chemicals added to serve as a control.

The results are shown in Fig. 3.

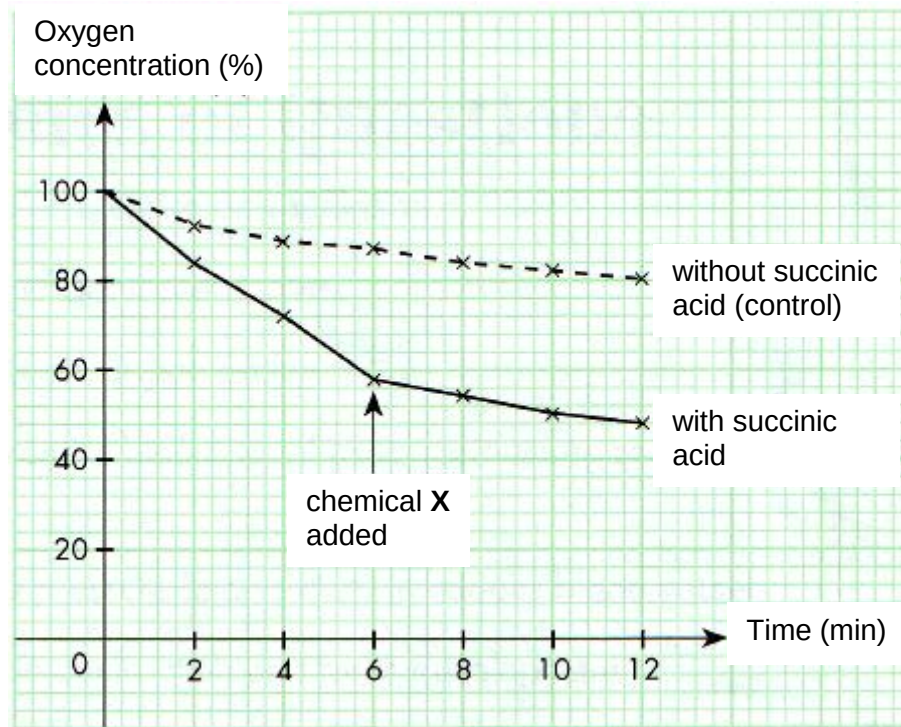


Fig. 3

- (a) Why was liver used to prepare the mitochondrial suspension but not other tissues? [2]

(b) Why are carbohydrate catabolism and oxygen consumption linked? [3]

(c) Account for the change in oxygen concentration in the control. [3]

(d) How are the changes in oxygen concentration in the reaction mixture different from those in the control from 0 to 6 minutes? [1]

- (e) Account for the difference mentioned in (d). [3]

- (f) State the effect of adding chemical X to the reaction mixture on the changes in oxygen concentration. Suggest a possible role of chemical X. [2]

[Total: 14]

- 4 On the Galapagos islands, there are 14 separate species of finch. When the finches' DNA was studied, it was found that all 14 species seemed to be more closely related to a single finch species in South America than to any other bird. It is suggested that a few of these South American finches were accidentally blown onto the islands at a time before there were other birds there.

Scientists have been studying finches on one of the Galapagos islands for over thirty years.

Before 1982, the only type of ground finch living on the island was the medium ground finch, *Geospiza fortis*.

In 1982, the large ground finch, *Geospiza magnirostris* settled on the island. The large ground finch is almost twice the size of the medium ground finch.

The key food for both species is the large seed from the plant *Tribulus cistoides*. Both species use their beaks to crack open the seeds.

During the period of the study, there have been two major droughts in 1977 and 2003. Some of the observations made by the scientists were listed below:

- The drought in 1977 wiped out mainly plant species producing small seeds, leaving *Tribulus cistoides* plants that produced large seeds with hard coats.
- Following the 1977 drought, the population of the medium ground finch was greatly reduced and the average beak size had increased in both length and depth by approximately 4%.
- The 2003 drought greatly reduced seed production by all plants including *Tribulus cistoides*.

Measurements were made of the mean beak lengths and mean beak depths of the medium ground finch population in both 2002 and 2005. These are shown in Table 4.1.

Table 4.1

	2002	2005
Mean beak length / mm	11.2	10.6
Mean beak depth / mm	9.4	8.6

- (a) State the **two** main selection pressures acting on the medium ground finch population following the 2003 drought. [2]

- (b)** Explain how the change in mean beak size in the medium ground finch population between 2002 and 2005 supports Darwin's theory of evolution. [4]

[Total: 6]

Section B**Answer EITHER 5 OR 6.**

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where applicable.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Either

- 5** **(a)** Describe the main structural features of endoplasmic reticulum and mitochondria. [7]
- (b)** Outline the functions of endoplasmic reticulum and mitochondria. [5]
- (c)** Explain how the structure of collagen is adapted to its function. [8]
- [Total: 20]

Or

- 6** **(a)** Compare the structure and function of *Taq* DNA polymerase and RNA polymerase. [7]
- (b)** Outline how restriction enzymes are used for removing sections of DNA from a chromosome. [5]
- (c)** Explain how gel electrophoresis can be used to provide evidence of molecular homology. [8]
- [Total: 20]